LUMA
A Benchmark Dataset for Learning from Uncertain and Multimodal Data

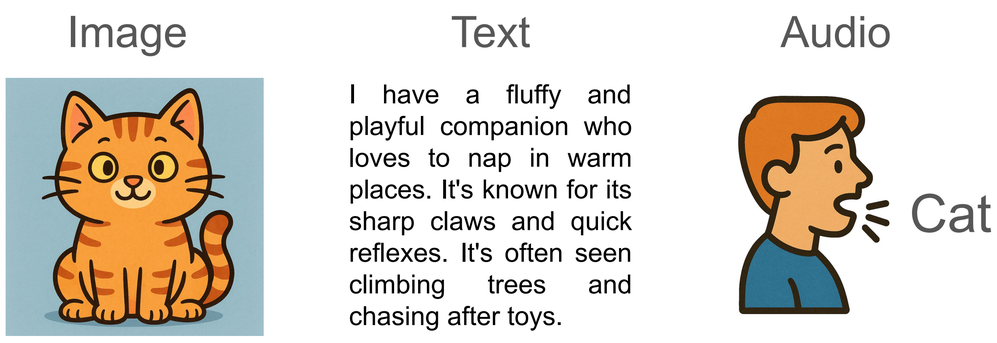
Grigor Bezirganyan[1] — grigor.bezirganyan@univ-amu.fr
Sana Sellami[1] — sana.sellami@univ-amu.fr
Laure Berti-Équille[2] — laure.berti@ird.fr
Sébastien Fournier[1] — sebastien.fournier@univ-amu.fr

[1]Aix-Marseille Univ, CNRS, LIS Marseille, France
[2]IRD, ESPACE-DEV Montpellier, France

Scan for Dataset

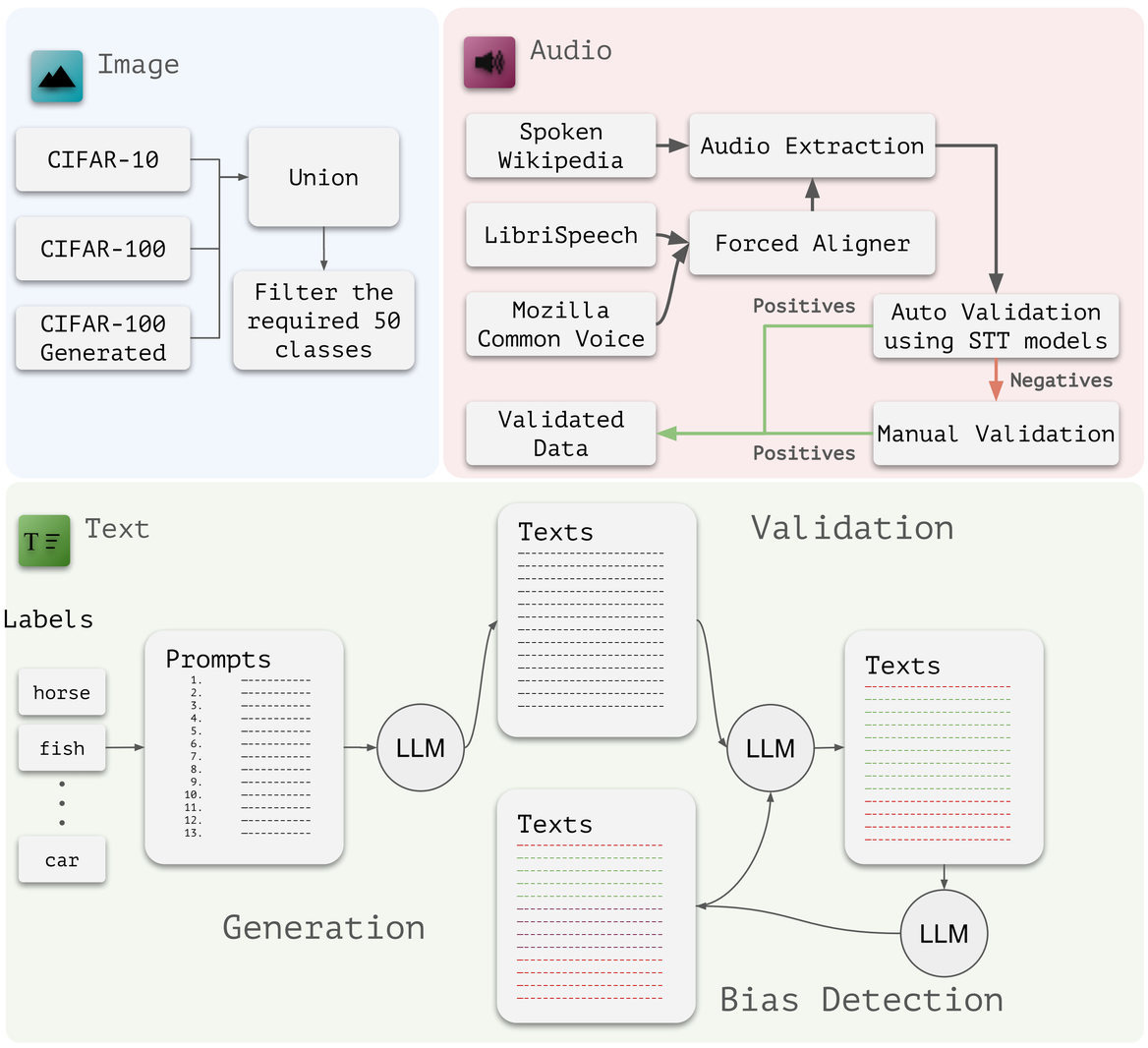
Context & Limitations

- **Multimodal AI** combines diverse sources for richer representations.
- **Uncertainty Quantification (UQ)** is key to evaluating model reliability.
- Most multimodal datasets are **not tailored for UQ experiments**.
- UQ benchmarks using simple noise (e.g., Gaussian) **don't capture real-world uncertainty**.
- Existing datasets are either **too simple for realism** or **too complex for controlled testing**.



Data Collection

- **Images** from CIFAR - 10 / 100 datasets.
- **Audio** samples forced aligned and extracted from Mozilla Common Voice, LibriSpeech, and The Spoken Wikipedia datasets
- **Text** samples generated using Gemma-7B LLM
- Bias & stereotype filtering on generated text samples
- Selected 42 classes + OOD classes



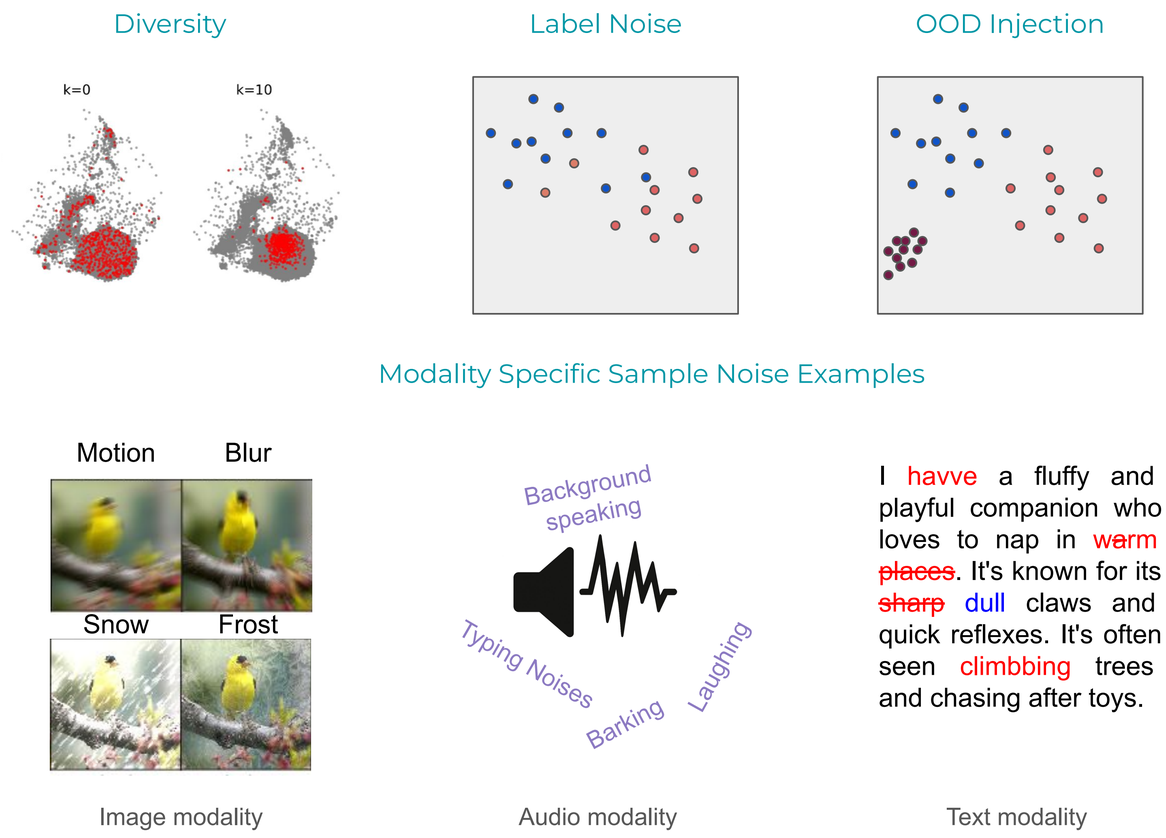
Further Reading

- Multimodal Learning with Uncertainty Quantification based on Discounted Belief Fusion (AISTATS 2025)
- EM-SEC: Efficient Multi-head Set-valued Evidential Classification (ECML-PKDD 2025)
- M2-Mixer: A Multimodal Mixer with Multi-head Loss for Classification from Multimodal Data (IEEE Big Data 2023)

Contributions

- A novel multimodal dataset **LUMA** designed to support research on learning under uncertainty in multimodal settings.
- Three modalities: **Image, Audio, and Text**, each describing the same semantic concept from different perspectives.
- A **Python package** for generating customized dataset versions with controlled noise injection per modality.
- Benchmarking **Multimodal Uncertainty Quantification** (MUQ).

Noise Injection



Modality Specific Sample Noise Examples

Image modality — Audio modality — Text modality

Uncertainty Quantification Baselines

Absolute values are reported for the clean LUMA dataset; noisy versions show percentage changes relative to it.

Method	Clean Dataset		↘ Diversity		↗Label Noise		↗Sample Noise	
	Ale.	Epi.	Ale. (%)	Epi.(%)	Ale.(%)	Epi.(%)	Ale.(%)	Epi.(%)
MCD Image	1.00	1.03	-15.73	-11.66	+59.20	+54.51	+4.44	+2.18
MCD Audio	0.52	0.70	-5.54	+2.16	+96.63	+54.49	+23.12	+14.40
MCD Text	0.37	1.01	-3.91	-2.62	+93.59	+2.41	+64.96	-2.03
MCD Multi.	0.26	0.78	-8.52	-1.21	+122.44	+11.60	+59.14	+9.89
DE Image	1.45	1.40	-37.49	-8.54	-7.43	+0.24	-18.46	-3.22
DE Audio	0.56	0.99	-27.39	-3.34	**+156.40**	+50.43	**+70.26**	+34.41
DE Text	0.42	1.01	+5.02	-6.15	+81.26	-0.51	+62.24	-7.11
DE Multi.	0.31	0.82	-22.80	-3.40	+115.15	+20.62	+45.97	+5.54
RCML Multi.	1.99	0.43	**+8.34**	**+16.16**	+64.72	**+106.16**	+36.19	**+58.21**

OOD Detection AUC Values for Different Methods

Method	MCD				DE				RCML Multi.
	Image	Audio	Text	Multi.	Image	Audio	Text	Multi.	
AUC	0.54	0.47	0.53	0.50	0.54	0.49	0.54	0.50	**0.91**

LUMA enables evaluation of MUQ methods under diverse noises and uncertainty types. LUMA further enables evaluation uncertainties on in-distribution and out-of-distribution data.